

Chronic Parkinsonism in humans due to a product of meperidine-analog synthesis.



Four persons developed marked parkinsonism after using an illicit drug intravenously. Analysis of the substance injected by two of these patients revealed primarily 1-methyl-4-phenyl-1,2,5,6-tetrahydropyridine (MPTP) with trace amounts of 1-methyl-4-phenyl-4-propionoxy-piperidine (MPPP). On the basis of the striking parkinsonian features observed in our patients, and additional pathological data from one previously reported case, it is proposed that this chemical selectively damages cells in the substantia nigra.